

AIRLINK AVIATION PVT. LTD.



DESIGN

We Design Inhouse with
Dedicated Engineer

Manufacturing

Design and Inhouse manufacturing
Capability of all Type of harness and
System Integration

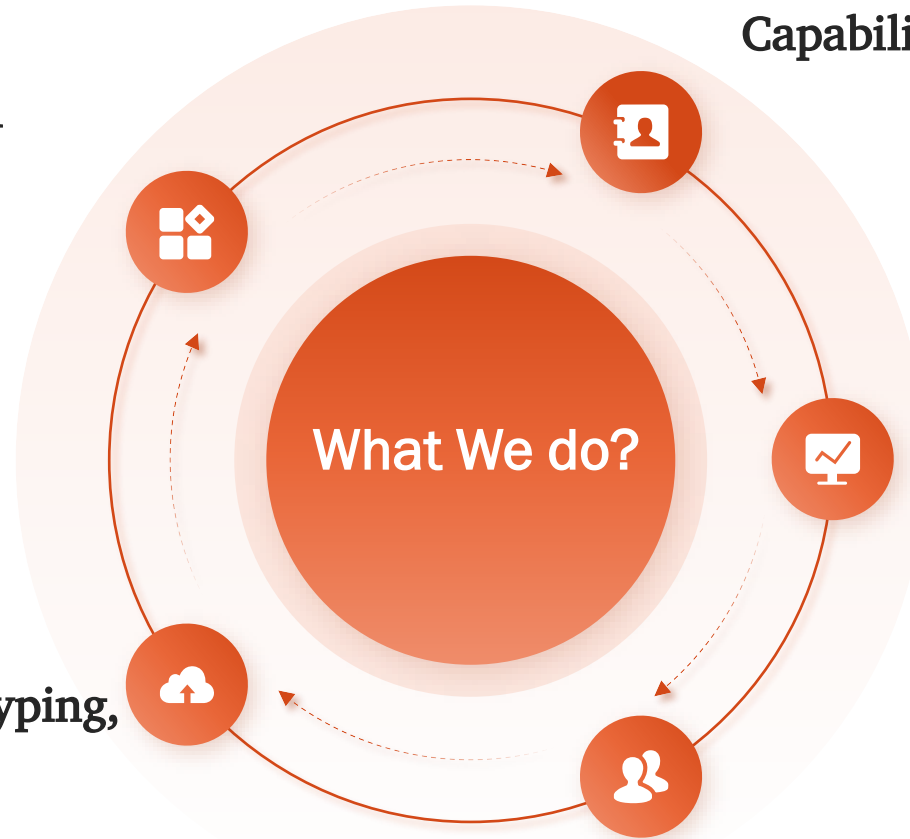
Built to Print

EMS - End-to-End solutions of
cable & wire harnesses,
electronic sub-systems

Built to Specification

Re-Design and Upgrade with More
efficient as per Needs

DEVELOPMENT
Product development, prototyping,
testing & qualifications



Our Services:



Simulator and Boards

We design and Manufacture Radio Simulator & Other Simulator Boards



Cable Harness Assembly

We do Simple, Complex, Multi brnch, EMI/EMC Protected and RF Cable Assemblies



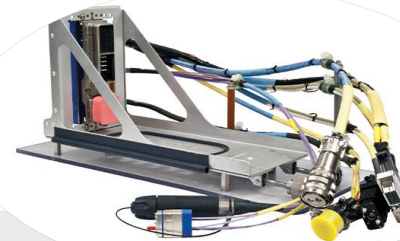
Multifunctional Displays

We design and Manufacture All Kind of Multi Functional Display and Console



Control Panels

Desing & Developoment of All Kind of control panels for Navy, Aerospace and Defence



Integration Assembly's

Panel Integration of Avionics, LRU,s, Chassis and Other Electronic Panels.



Market and Application



Aerospace



Defense



UAV



EV

Explore Our Solutions

Radio Simulators and Boards

01

16 Chanel Radio Simulatio and Simulation
Design, Devolopment and Fabrication

Wire Harness Solutions

02

Built to Print, Built to Specification, Customised,
EMI/EMC, Complex Harness and RF Solutions

Multifunctional Displays

03

Multifunctional Display for Avionics, Navy
and For Ground Systems

Control Panels

04

Control Panel Design and Integration for
Aeronautics, Tanks, Defence and Navy Application

Integration of Panels

05

Integration of Panel as per the spec, Design as per
the need, Devolopment of PDU's and PCB's

Other Assemblies & Source of BOM Parts

06

We support for other Assemblies and we Support
for Sourcing for any other BOM Parts.



Radio Simulator

Radio Simulator Boards are hardware platforms designed to simulate and route RF signals between multiple radios or communication ports. The matrix configuration determines how many inputs (Tx/Rx ports) and outputs (antenna paths or radios) can be interconnected.



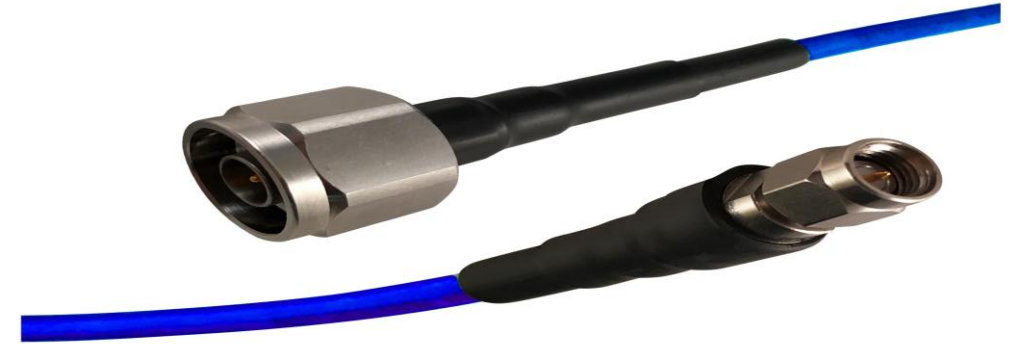
SPECIFICATION:

- (6X6) & (4X4) channel radio simulator with mil grade circular connector,
- 10 TOS station and 10 IOS station configuration for cross talk with PTT



Cable Harness Solution

In House Harness capability



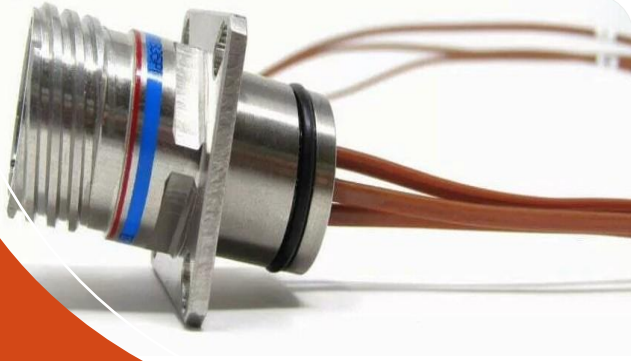
We offer Custom Harness Solutions of any make and any OEMs with standard MIL cable and specific cables and connector harness solution.

Our capabilities include Electrical Wiring Interconnection Systems, High Precision Electromechanical Systems, and Mission Critical Electronic Control Systems.

We also specialize in Specialty Products – RF and Flat Cable Interconnections, and provide Rugged Compact Systems for Airborne Applications, built to meet stringent environmental and operational standards.



Cable Harness Solution



Simple Harness

- Pig Tail Harness.
- Back to Back Harbess
- Non-EMI/EMC Harness
- Lacing and Bunduling



Medium Harness

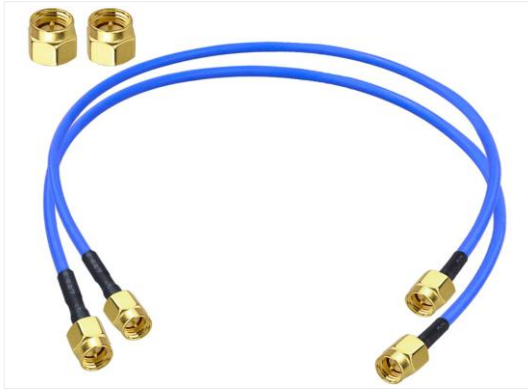
- Connecting with Backshell
- Metal Braiding
- Backpotting solution
- Soldering Process



Complex Harness

- Multiple Branch Harness with Routings.
- 1:1 Drawing creation & Used in Routings.
- Big Looms with all type of Connector along with Soldering and Critical Process.

RF Cable Solution



Coaxial cable assembly used to link devices with different connectors

High-frequency SMA (SubMiniature version A) for compact gear like Wi-Fi with rugged, high-power N-Type for antennas or base stations, offering 50Ω impedance, durability, and signal integrity from DC up to several GHz



We Can Connect with Different Size connector

Like SMA to SMA, SMA to N-Type, N-Type to N-Type, BNC Assembly, MMCX, MCX, TNC and Other connection of RF Connectors.

50Ω & 75Ω Cable harness.



We do with Semi Rigid and Rigid RF Cables Harness

Uses coaxial cables with solid metal jackets (copper/aluminum) for excellent EMI shielding, phase stability, and low loss, ideal for high-frequency (DC to 67+ GHz) aerospace, defense, and telecom applications

Connector/termini of fiber optics

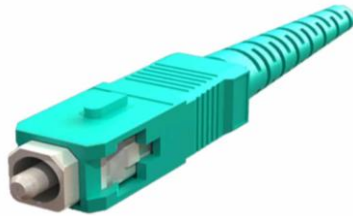
- **Bayonets**
 - ST series



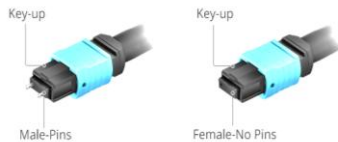
- **Screw-in**
 - FC series



- **Push-pull snap-in**
 - SC, E C Series
 - MTP/MPO



- **Push-pull latched (RJ45)**
 - LC Series



Single-Fiber contact

- Only one fiber per contact
- Easy termination
- Cost effective
- Easy maintenance

- **D3899 connector**
 - Arinc-801 termini



- **Expanded beam**
 - EB termini



- **MT Expanded beam**
 - MT EB ferrule



- Colour coding is the same for the adapters and for contact :
 - **Blue: Single-Mode PC**
 - **Green: Single-Mode APC**
 - **Beige: Multimode PC**
 - **Aqua: Multimode PC with OM3 or OM4 Fiber**

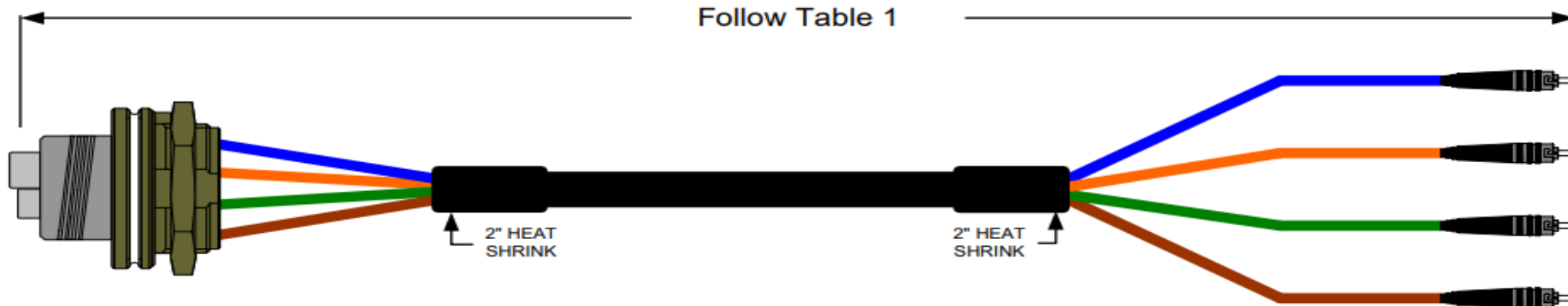
MT Ferrule

- From 12 to 72 fibers inside one ferrule
- High density of fiber
- Very compact solution
- High cost
- Complex termination

FIBER OPTIC CABLE ASSEMBLY

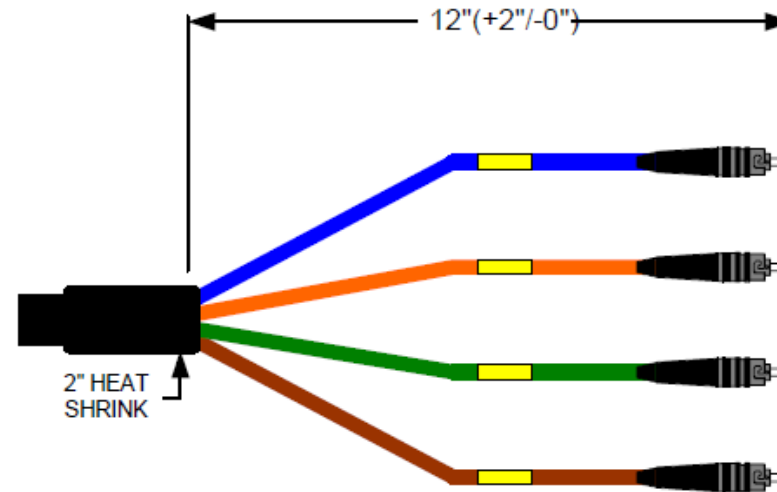
Multi-channel connector (plug/receptacle) to pig-tail (fan-out)

Front face of the insert cap on multi-channel connector to ferrule tips of individual connectors.



Length of Breakouts (fan-out)

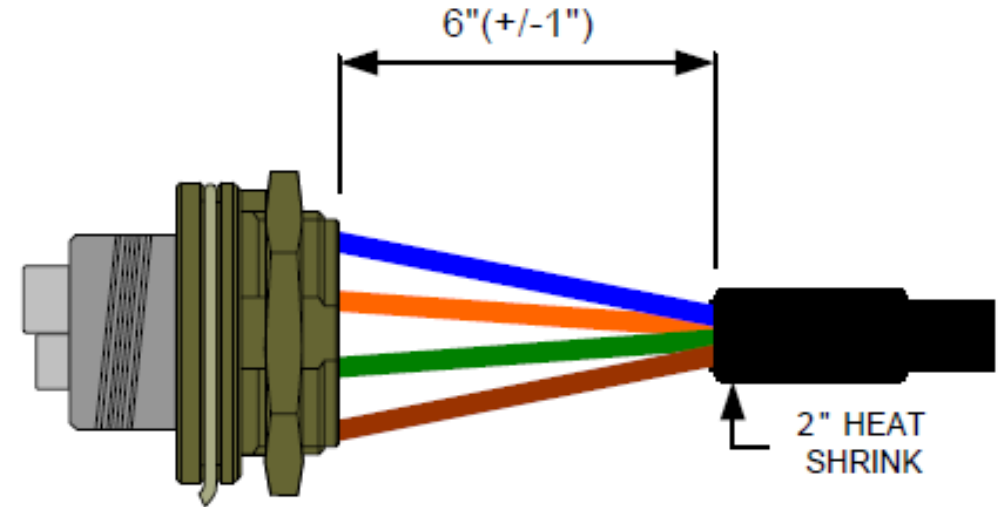
- Length: 12 inches is typical but it can be specified.
- Tolerance: +2.0 inches / -0.0 inch is typical. If the pig-tail length exceeds the typical length, use the tolerances in Table 1.



FIBER OPTIC CABLE ASSEMBLY

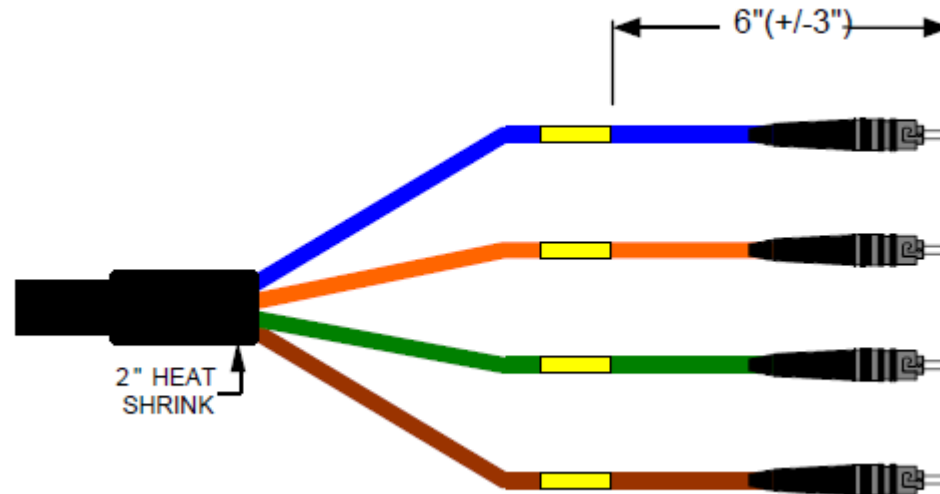
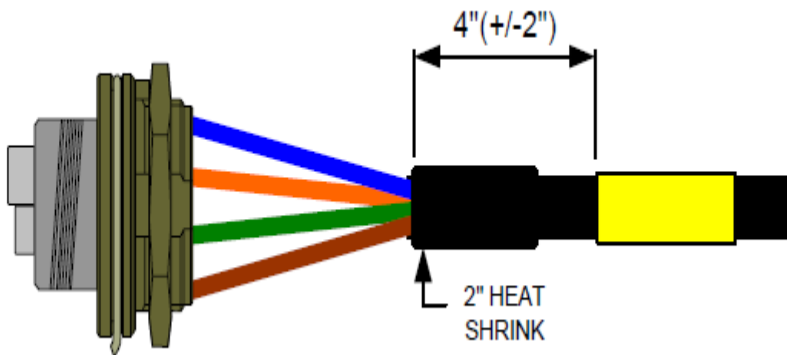
Length of Breakout Behind Plug/Receptacle

- Length: 6 inches is standard.
- Tolerance: +/- 1.0 inch is typical.
- Dimension lines shall be placed at the edge of the heat shrink.



LABELING REQUIREMENTS:

- Heat shrink tubing - Character size, Number of lines of information Orientation and Location



Multifunctional Display

01

Multifunction Displays (MFDs) are versatile electronic screens that integrate multiple functions, like navigation, communication, and system monitoring, into a single interface common in aviation (glass cockpits), marine electronics, and industrial controls, replacing dedicated gauges with configurable digital displays for weather, radar, sonar, or engine data

Integrated Information: Combines data from various sensors and systems (GPS, radar, sonar, engines, autopilot) onto one screen.

Configurable Views: Users can customize screens to show different data pages (e.g., moving maps, system status, weather).

02



Control Panels

Capable to Design, Development and Manufacturing of Control Panels

Defining system needs, creating detailed electrical schematics and layouts (using software like SolidWorks), selecting components (PLCs, HMIs, contactors, breakers), assembling

hardware in a protective enclosure (steel, aluminum, plastic), and rigorous testing to ensure reliable automation for Systems, requiring expertise in electrical, mechanical, and integration for safety and performance.

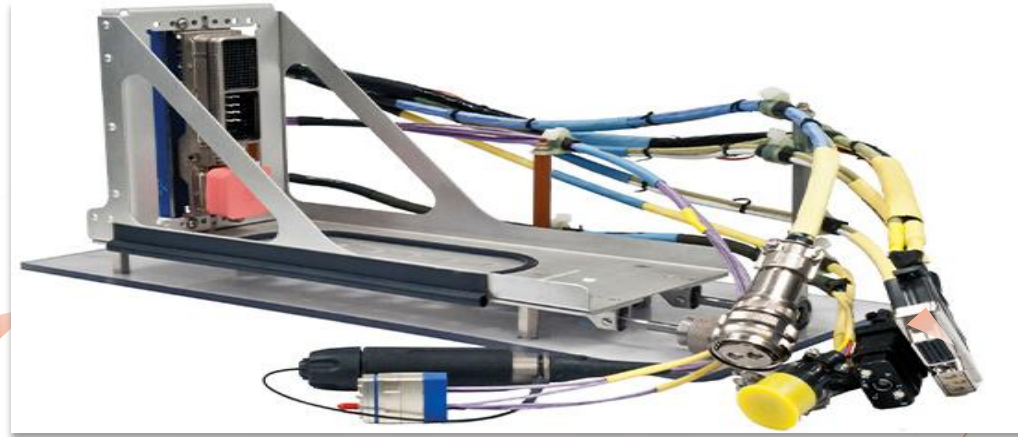


Integration of System



Design and Development

We design and Develop the Electronic Modules Like LRU's, PDU System and Others



Built To Specification

Manufacturing model where the customer provides performance and functional requirements, and the manufacturer handles the entire design, engineering, production, and assembly of a finished electronic product



Built To Print

Manufacturing model where a provider builds complex electronic systems, components, or assemblies precisely following a client's detailed designs, drawings and specifications

Sourcing and Distribution

Connectors

We source the Varieties of
connector directly from OEM
and Re-sell you

Cables

We supply Different Make and
Different type of cables to
DRDO Labs

Other Readymade solutions

We support to in Readymade solution
like Electronic components,
Softwares and Work stations



Airlink Aviation Pvt Ltd.



**WORKS FOR DEFENSE, AEROSPACE,
NAVAL AND MILITARY APPLICATIONS**

